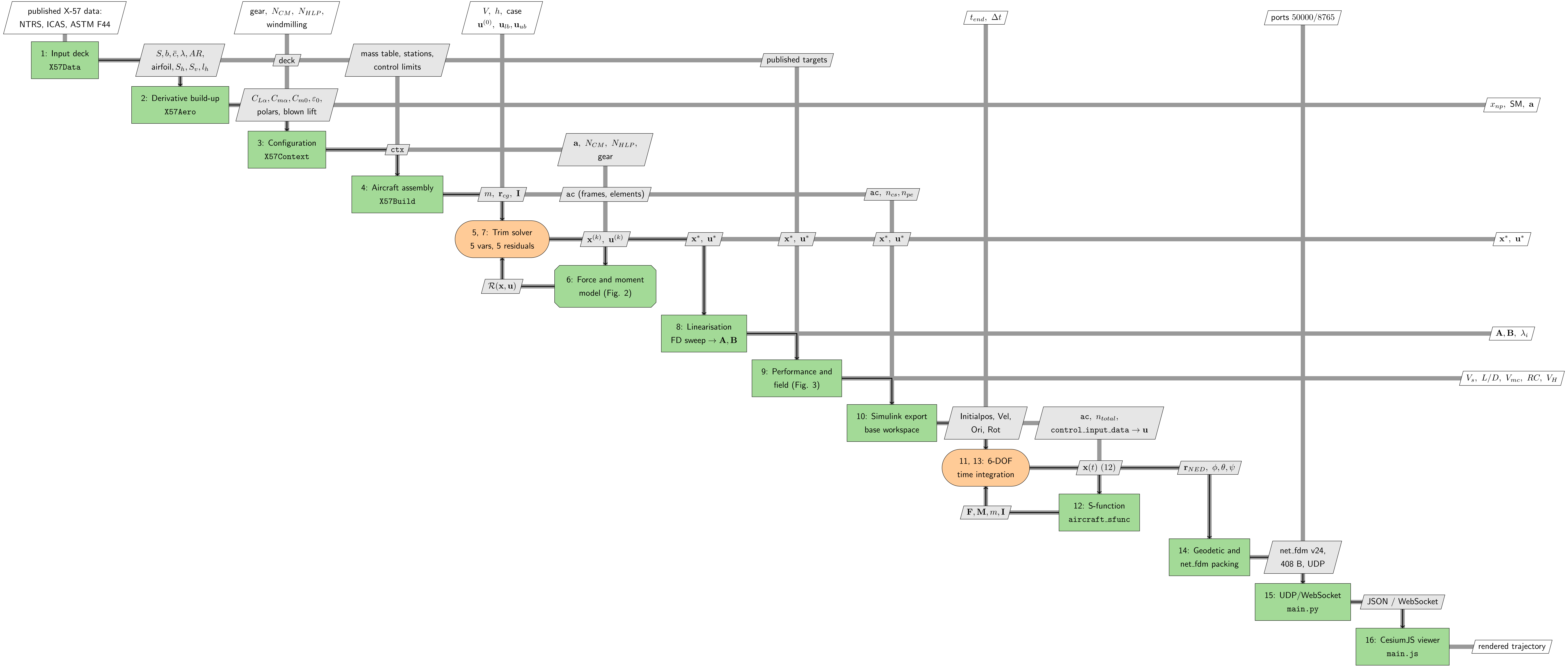
published X-57 data: NTRS, ICAS, ASTM F44
gear, N_{CM}, N_{HLP}, windmilling
V, h, case $\mathbf{u}^{(0)}$, $\mathbf{u}_{lb}, \mathbf{u}_{ub}$
t_{end}, Δt
ports 50000/8765
1: Input deck X57Data
$S, b, \bar{c}, \lambda, AR$, airfoil, S_h, S_v, l_h
deck
mass table, stations, control limits
published targets
2: Derivative build-up X57Aero
$C_{L\alpha}, C_{m\alpha}, C_{m0}, \varepsilon_0$, polars, blown lift
x_{np}, SM, $\mathbf{a}$
3: Configuration X57Context
ctx
$\mathbf{a}$, N_{CM}, N_{HLP}, gear
4: Aircraft assembly X57Build
m, $\mathbf{r}_{cg}$, $\mathbf{I}$
ac (frames, elements)
ac, n_{cs}, n_{pe}
5, 7: Trim solver 5 vars, 5 residuals
$\mathbf{x}^{(k)}$, $\mathbf{u}^{(k)}$
$\mathbf{x}^*$, $\mathbf{u}^*$
$\mathbf{x}^*$, $\mathbf{u}^*$
$\mathbf{x}^*$, $\mathbf{u}^*$
$\mathbf{x}^*$, $\mathbf{u}^*$
$\mathcal{R}(\mathbf{x}, \mathbf{u})$
6: Force and moment model (Fig. 2)
8: Linearisation FD sweep → $\mathbf{A}, \mathbf{B}$
$\mathbf{A}, \mathbf{B}$, λ_i
9: Performance and field (Fig. 3)
V_s, L/D, V_{mc}, RC, V_H
10: Simulink export base workspace
Initialpos, Vel, Ori, Rot
ac, n_{total}, control_input_data → $\mathbf{u}$
11, 13: 6-DOF time integration
$\mathbf{x}(t)$ (12)
$\mathbf{r}_{NED}$, ϕ, θ, ψ
$\mathbf{F}, \mathbf{M}, m, \mathbf{I}$
12: S-function aircraft_sfunc
14: Geodetic and net_fdm packing
net_fdm v24, 408 B, UDP
15: UDP/WebSocket main.py
JSON / WebSocket
16: CesiumJS viewer main.js
rendered trajectory